Optical-Enhanced Oil Tank Detection in High-Resolution SAR Images

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***Abstract*— In recent years, object detection in high-resolution synthetic aperture radar (SAR) images has made significant progress, especially after the introduction of deep learning. However, objects such as dense oil tanks, which are compactly arranged in SAR images, are still challenging to recognize due to the unique imaging mechanism of SAR. Inspired by human learning from comparison, we propose a multistage framework for oil tank detection in SAR images using optical image enhancement. Specifically, in the training stage, we build a teacher–student network to align the semantic information between the two modalities, where the optical features are used to guide the corresponding SAR feature learning, while in the inference stage, the learned network detects oil tanks using only SAR images as input. Besides, a pretraining stage before training is applied to further improve the network's ability for SAR feature extraction, which is realized by the proposed paired optical-SAR self-supervised learning. To verify the effectiveness of the proposed method, we perform experiments on our newly built SpaceNet6-OTD dataset. Extensive experiments demonstrate that the proposed method can effectively improve the accuracy of detecting oil tanks in SAR images. Datasets, codes, and more results will be released at: https://EIS-VIPG.github.io/SpaceNet6-OTD/.**

***Index Terms*— Knowledge distillation (KD), oil tank detection, self-supervised learning (SSL), synthetic aperture radar (SAR) image.**

I. Introduction

SYNTHETIC aperture radar (SAR) sensors have the unique capability to penetrate clouds and collect during all-weather, day, and night conditions. It has been widely used in both civilian and military fields, such as disaster response [1], marine observation [2], resource exploration [3], crop yield estimation [4], and mapping [5]. High-resolution SAR data can provide more valuable information when weather and cloud cover obstruct traditional optical sensors, which are suitable for detecting and monitoring some typical objects, such as buildings, ships, planes, vehicles, and oil tanks. In this work, we focus on detecting oil tanks in high-resolution SAR images, which is very challenging due to the unique imaging mechanism of SAR images.

The imaging mechanism of SAR is fundamentally different from that of optical remote sensing and the perception of images of human vision. Therefore, the objects' appearances presented in SAR images are not only completely different from those familiar to our visual system but also very different from the optical remote sensing images. SAR sensors image by actively illuminating the ground objects with radio waves and receiving the backscattering energy formed by each independent scattering unit of the corresponding object. Thus, the pixel value in the SAR image represents how much radar energy is reflected by the sensor and the boundaries of ground objects in SAR images mainly manifest as a discrete combination of bright points, lines, and blobs, with poor continuity, integrity, and severe geometric deformation. The oil tanks in SAR images have more typical appearances. They often present in clusters, with very different density distributions. Fig. 1 shows several SAR oil tank images with different density distributions. In the sparse case, it can be observed that the roof of an oil tank target appears as a bright ring in the SAR image, while its bottom appears as a black shadow accompanied by a bright arc with a strong scattering center. In the dense case, a close distance between the oil tanks causes the severe overlap of the bright areas, and the shadow areas are also occluded, which makes it more difficult to distinguish the different oil tank instances. In the very dense case, we can only observe a bright area, there is serious scattering interference between the grouped tanks, and even the bright ring or arc appearances have been lost. Based on the above, the blurred and cluttered appearances of the dense oil tanks in high-resolution SAR images are very hard to distinguish even for the experts of SAR image analysis.

Traditional oil tank detection methods often apply saliency analysis or circular detection based on the geometric shape information [7], [8], [9], which is easily affected by other objects with similar shapes and does not take fully the scattering interference between oil tanks into account. Recently, benefitting from the powerful image representation performance of deep learning, plenty of methods have used deep neural networks to detect an object in remote sensing images

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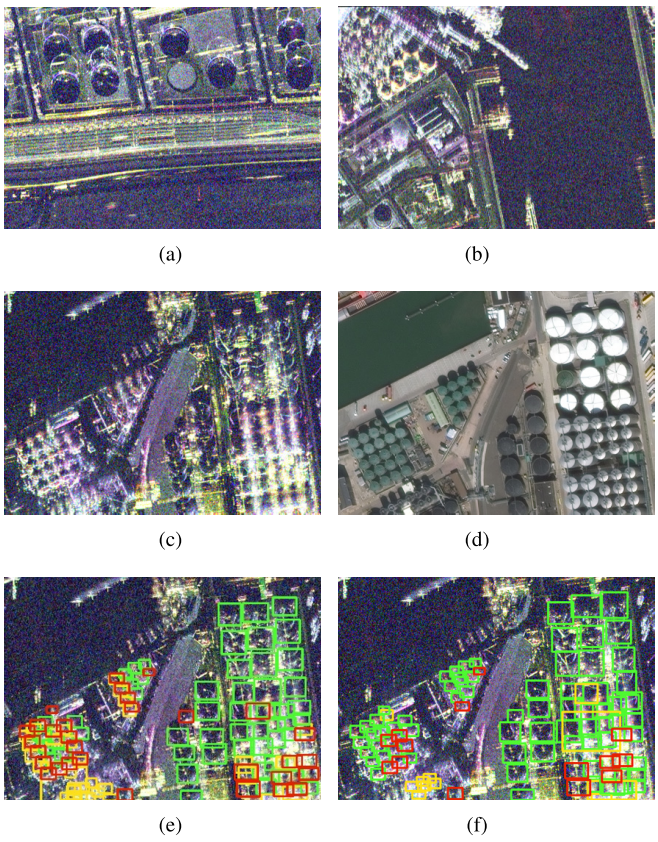


Fig. 1. Oil tanks of varying density in high-resolution SAR images. The SAR intensity is visualized with three polarizations (HH, VV, and HV) displayed through the red, green, and blue channels. (a) Sparse oil tanks (SAR). (b) Dense oil tanks (SAR). (c) Very dense oil tanks (SAR). (d) Corresponding optical image of very dense oil tanks. (e) Detection result of DetectoRS. (f) Detection result of our method. (The green boxes refer to the correct detection results, the yellow boxes refer to the false detection results, and the red boxes refer to the missing targets.)

and have achieved the state-of-the-art accuracy [10], [11], [12], [13]. However, although these methods consider some characteristics of remote sensing images, such as rotation invariance, they cannot well handle the densely arranged oil tanks in SAR images due to the vast appearance differences. Ma et al. [14] introduced several hand-designed modules into the convolution neural network combining with the scattering properties, which provides a new way to detect oil tanks in SAR images. However, SAR imaging parameters are needed, which are not always available.

Though some existing methods can accurately detect sparse oil tanks with clear shapes, they suffer from great difficulties in detecting oil tanks in high-resolution SAR images due to the indistinguishable appearance, especially the dense oil tanks. A natural question is how can we teach the network learning to recognize the oil tanks in high-resolution SAR images? As we all know, when humans encounter an illegible image of an unfamiliar modality, we often learn to recognize it by comparing it with the corresponding image of a recognizable modality. Inspired by that, we propose to introduce optical images that are more suitable for human vision to guide the network in learning the feature of oil tanks in SAR images. Specifically, from the perspective of feature extractor in deep

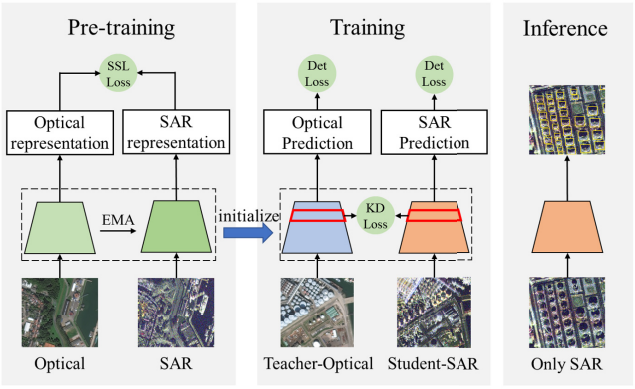


Fig. 2. Proposed pipeline for detecting oil tanks in SAR images, using OS-SSL to pretrain a better feature extractor as the initialization in the training stage. First, the optical network guides the SAR network as the teacher. Then, the model detects the oil tanks during inference using only SAR images as input.

learning, we introduce optical images to optimize the SAR feature extractor. Deep learning-based methods usually require numerous labeled data for training to obtain a satisfactory model. However, the high labeling cost makes it impossible for supervised pretraining using large amounts of SAR images. When the performance deteriorates dramatically due to the limited training samples, we often load the model pretrained on ImageNet for better initialization. Still, it is not the best for SAR feature extraction because of the modality differences. Therefore, we use self-supervised learning (SSL) to pretrain a feature extractor more suitable for SAR images. Meanwhile, we introduce the optical images to further improve the network's learning ability for SAR images. In addition to the image-level pretraining, we also employ the information of oil tanks in optical images to guide the network to learn better SAR features by utilizing cross-modal knowledge distillation (KD) in an object-level way.

In this article, we design a multistage pipeline for detecting oil tanks in SAR images with the enhancement of optical images, as shown in Fig. 2. First, optical-guided SSL is used for the image-level pretraining to obtain the feature extractor suitable for SAR images. Then, in the training stage, a teacher–student network is constructed to make the SAR branch learn how to recognize oil tanks from the optical branch through KD, where the guidance of optical image information can be seen as object-level fine-tuning for SAR feature learning. Finally, the trained model detects the oil tanks using only SAR images as input during the inference stage. Our contribution can be summarized as follows.

1) We propose a pipeline that uses optical images to guide the network in training a better oil tank detector for SAR images. Co-registered optical and SAR images are used in the training stage, while only SAR images are used in the inference stage.
2) We introduce paired optical-SAR images into the SSL to obtain a better SAR image feature extractor, improving the performance of detecting oil tanks in SAR images by 8%–9%.

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3) We construct an oil tank detection dataset, SpaceNet6-OTD, to validate the effectiveness of the proposed method. The dataset contains carefully labeled 820 SAR-Optical images with 900 × 900 pixels.

The rest of this article is organized as follows. In Section II, we briefly describe the related works. A detailed description of the proposed method, including the optical-SAR self-supervised pretraining and the teacher–student detector, is then provided in Section III. The experimental results are discussed in Section IV. Finally, we draw a conclusion in Section V.

II. Related Work

In this section, we first review the recent progress of oil tank detection in SAR images. Next, we describe the two main components specifically related to our design: SSL and cross-modal KD.

A. Oil Tank Detection in Remote Sensing Images

Traditional oil tank detection methods often use hand-designed features and are unsupervised. One category of typical methods uses geometric shape information of oil tanks. Xu et al. [15] combined the quasi-circular shadows and highlighting arcs to detect oil tanks in SAR images. Liu et al. [16] built a circular feature map based on a radial symmetric circle and then combined the latent saliency map with the circular feature map to detect the oil tank in optical images. Another category is based on the saliency analysis. Zhang et al. [17] and Zhang and Liu [18], [19] proposed several methods based on a saliency analysis for oil tank detection in SAR images. In addition, Wang et al. [20] fused the information of optical images and SAR images to detect the oil tank. The main limitations of these approaches are the weak applicability and unsatisfactory detection accuracy for high-resolution SAR images.

Recently, more and more object detection methods based on deep learning emerge and push the performance to a new height. Some pioneer methods, such as Faster R-CNN [21], Cascade R-CNN [22], and DertectoRS [6], can achieve fantastic performance, which can also be used for high-precision oil tank detection. For remote sensing images, according to the general object characteristics, some studies introduce the rotation invariance into the detector to achieve oriented object detection [10], [11], [12]. However, most of these methods are designed for optical remote sensing images, and the rotation invariance is not necessary for oil tanks. A few works use deep learning to solve the oil tank detection in SAR images. For example, Ma et al. [14] introduced a transformer network together with prior sensor information, realizing the detection and 3-D modeling of oil tanks. These methods detect oil tanks mainly based on the texture and structural features, but the representation of this information in SAR images is weak and hard to extract. Our method introduces optical images to assist the SAR image feature learning, helping the network extract better SAR features for detection.

B. Self-Supervised Learning

SSL aims to use unlabeled data to learn image representations that are transferable to downstream tasks. Early studies typically focus on solving hand-designed "pretext task" such as patch sorting [23], [24], inpainting [25], colorization [26], and rotation prediction [27]. In recent years, since the proposal of InfoNCE [28], many works based on contrastive learning have pushed the performance to a new height, such as MoCo [29] and SimCLR [30]. They learn image representations by training networks to distinguish positive and negative samples. Some later works accomplish SSL with only positive samples. BYOL [31] constructs an asymmetric architecture and trains through the mutual prediction between positive sample pairs. To prevent collapse in the training, the weight of one branch is used as the momentum encoder of another. The above methods mainly focus on object-centric natural images but do not work well in remote sensing images.

While SSL has been extensively studied on natural images, its research in remote sensing domain is still insufficient. Early study [32] uses small datasets specific to a small geographical region, which limits the generalization. Recently, contrastive learning brings great improvement to SSL, and some similar methods are also extended to the field of remote sensing images. Uzkent et al. [33] proposed to learn the representation of satellite images by predicting the information extracted from Wikipedia articles of the corresponding locations. Mañas et al. [34] and Ayush et al. [35] used multitemporal data to construct self-supervised positive and negative samples. According to the pioneer studies, it is evident that the core problem of contrastive SSL for remote sensing images is the construction of positive or negative pairs. Our method takes paired optical-SAR images as positive pairs and uses the differences between modalities as natural augmentation, which can avoid the scene changes due to some inappropriate augmentations. Meanwhile, the introduction of optical images brings richer information, further easing the limitation on the number of samples.

C. Cross-Modal KD

Early KD aims to build a smaller model with the softmax labels of a larger model [36]. Komodakis and Zagoruyko [37] forced the student network to mimic middle-level attention maps of the teacher network besides its output probability maps. KD in object detection usually adopts feature-based distillation. Early methods [38], [39], [40], [41] use the features of the objects and their context as the distillation target, where the proposals or region of interests (RoI) are often used to determine the target locations. Recent methods [42], [43], [44], [45] hold the view that all different spatial locations of feature maps contain useful information and achieve better distillation effects by adjusting the weights of different locations of the feature maps. However, these KD methods usually focus on single-modality data and cannot adapt well to cross-modal tasks.

Recently, some researchers have extended it to cross-modal learning. Cross-modal KD aims to transfer knowledge across different modalities. Many of them rely on paired data with the same labels and directly apply KD with features extracted from different modalities, where features extracted by the teacher network are used to supervise the student network.

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This distillation method works well for a variety of cross-modal tasks, such as between text and image [46], infrared and RGB images [47], optical and SAR images [48], [49], and depth and RGB images [50], [51]. There are also some methods using the probability distribution of output space when processing unpaired samples from different modalities [52]. Some works [53], [54], [55], [56] generate modal-like images through the generative adversarial network, and then, the Kullback–Leibler divergence loss is calculated by using the output softmax results to achieve KD. Our goal is to use the corresponding optical images to assist SAR images for oil tank detection, and thus, distillation in feature space is more direct and effective. Two closely related methods to ours are the work of Kim et al. [48] and Kang et al. [49], and Kim et al. [48] used optical features to supervise SAR features directly. However, optical images have different information from SAR images, and it is unreasonable to force SAR features to learn optical features directly. Kang et al. [49] constructed two flows in the deep stage of networks and supervised the optical semantic flow with pretrained optical features, which can alleviate this problem to some extent. Our method utilizes the attention map to transfer the knowledge, and the semantics of the deep representations is identical between the two modalities and is more suitable for supervision.

III. Methods

This section details our proposed pipeline, including the optical-SAR SSL (OS-SSL) for the pretraining stage and the teacher–student detector based on KD for the training stage.

A. Optical-SAR SSL

Mainstream contrastive SSL approaches, such as MoCo [29] and SimCLR [30], learn image representations by maximizing the similarity between positive samples, where the positive samples are produced by applying different augmentation to the same image. Also, many negative examples are used for comparing to avoid the collapse in training. However, considering that there are many similar ground objects in remote sensing images, such as buildings and roads, the negative samples randomly selected may have certain similarities with the positive samples, which are not suitable for comparison. Therefore, we focus on the state-of-the-art contrastive SSL framework BYOL and predict in representation space with only positive samples. At the same time, our goal is to introduce optical images to enhance the representations extracted from SAR images. Therefore, we improve the method for constructing positive pairs tailored to our OS-SSL. Specifically, paired SAR and optical images are input into the network as positive sample pairs, and the obtained SAR features and optical features are predicted mutually. In this way, the feature extractor can extract the SAR features closer to the optical features.

We use two neural network branches with different parameters θ and ξ, as shown in Fig. 3. They both have an encoder f and a projector g; besides, one branch contains a predictor q that predicts the regression targets of the other branch. In training, one branch does not have the backward propagation but updates the parameters through the exponential moving average (EMA) of the other branch's parameters. More precisely, given a decay rate $\tau \in [0, 1]$, after each training step, we perform the following update:

$$\xi \leftarrow \tau\xi + (1-\tau)\theta. \tag{1}$$

Let x^{sar} and x^{opt} represent a set X consisting of the same scene's SAR and optical image, respectively. Paired x^{sar} and x^{opt} are fed into the encoder of online and target network, respectively. For SAR images, the network outputs an SAR representation $y_\theta^{\text{sar}} = f_\theta(x^{\text{sar}})$ and an SAR projection $z_\theta^{\text{sar}} = g_\theta(y_\theta^{\text{sar}})$. Another network outputs an optical representation $y_\xi^{\text{opt}} = f_\xi(x^{\text{opt}})$ and an optical projection $z_\xi^{\text{opt}} = g_\xi(y_\xi^{\text{opt}})$ for the optical images. We then output the SAR prediction $q_\theta(z_\theta^{\text{sar}})$ to regress the optical projection z_ξ^{opt}. Note that the predictor here makes the architecture asymmetric between the two branches, which can avoid the collapse in training. Finally, we define the following mean squared errors between the SAR predictions and optical projections:

$$L_{\theta,\xi} = \left\| \bar{q}_\theta\left(z_\theta^{\text{sar}}\right) - \bar{z}_\xi^{\text{opt}} \right\|_2^2 \tag{2}$$

where $\bar{q}_\theta(z_\theta^{\text{sar}})$ and $\bar{z}_\xi^{\text{opt}}$ mean the ℓ_2-normalization of $q_\theta(z_\theta^{\text{sar}})$ and z_ξ^{opt}, respectively.

According to the above, the networks learn to use SAR representation to predict optical representation. Then, we symmetrize the loss $L_{\theta,\xi}$ in (2) by swapping inputs to compute $\widetilde{\mathcal{L}}_{\theta,\xi}$ and use optical representation to predict SAR representation. The mutual prediction can narrow the distance between the optical feature space and the SAR feature space and further optimize the extracted SAR features. Finally, the parameter θ is updated by minimizing the SSL loss function

$$L^{\text{self}} = L_{\theta,\xi} + \widetilde{\mathcal{L}}_{\theta,\xi}. \tag{3}$$

B. Teacher–Student Detector Based on KD

Compared with SAR images, optical images are better for object discrimination and can provide favorable prior knowledge for understanding SAR images. Our goal is to improve the learning process by transferring knowledge from the optical to SAR. The transferred knowledge is distilled by aligning the semantics of the deep representations and can be viewed as extra supervision in addition to the ground truth. To this end, we build a teacher–student network to realize the KD and use the detection model training with optical images as the teacher, while the SAR branch is the student.

We use the same two-stage detector Faster R-CNN [21] for the teacher network and the student network, as shown in Fig. 4. The basic detector consists of backbone, region proposal network (RPN), and regression head. Specifically, we used ResNet50 [57] to extract features and build a feature pyramid with FPN [58] in the backbone. The four stages $\{C_2, C_3, C_4, C_5\}$ in ResNet50 are used to generate the different levels' features $\{P_2, P_3, P_4, P_5, P_6\}$ of the feature pyramid by a top-down connection. The higher the level of the feature, the smaller the feature size, and the larger

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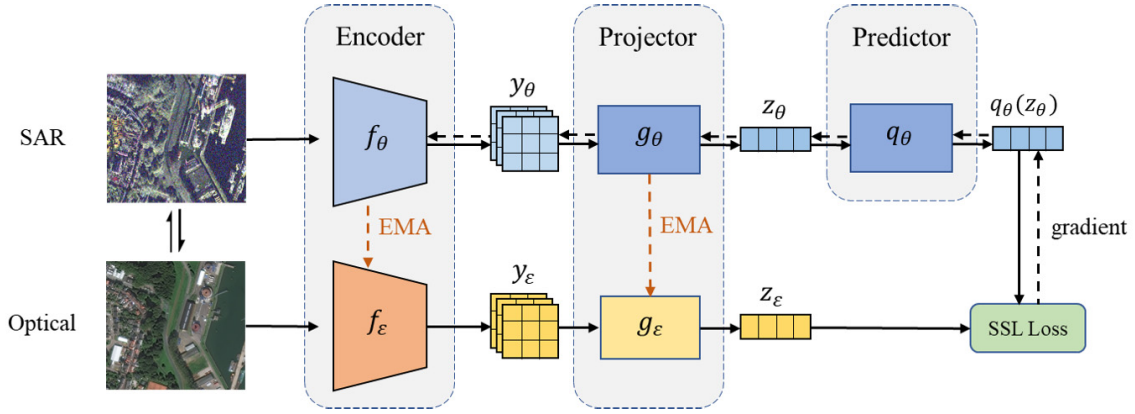


Fig. 3. Architecture of OS-SSL network. θ is the trained weight and updated by the backpropagation, and ξ is the EMA of θ. At the end of training, f_θ is used as the pretrained feature extractor.

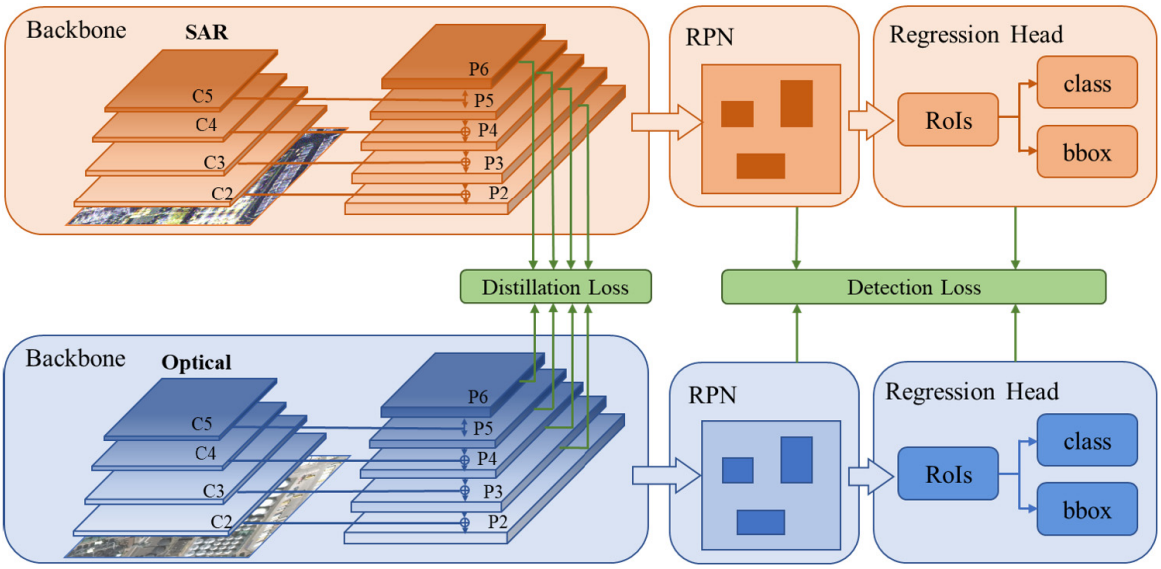


Fig. 4. Architecture of KD-based teacher–student detector. Similar Faster R-CNN detectors are used as the teacher and student networks. The teacher network is trained with the optical images in advance, and the student network is trained with the supervision of SAR oil tank ground truth and the knowledge of the optical features.

reception field is conducive to feature representation. RPN preliminarily predicts the foreground and background and regresses the rough proposal boxes by laying anchors on $\{P_2, P_3, P_4, P_5, P_6\}$. Regression head translates the features of proposals into 7×7 feature patches as RoIs and then predicts the classification confidence and the coordinates of bounding box for each RoI. The loss of the detector L_{bbox} is calculated by

$$L_{\text{det}} = L_{\text{RPN}} + L_{\text{cls}} + L_{\text{bbox}} \tag{4}$$

where L_{RPN}, L_{cls}, and L_{bbox} represent the RPN loss, object classification cross entropy (CE) loss, and the bounding bbox regression Smooth L_1 loss, respectively [21].

According to semantic consistency, the activation distribution of oil tanks on the feature maps should be consistent. Thus, we distill the knowledge of optical branch by aligning the activation distribution of the two modalities. Specifically, we calculate the attention map of the feature map to represent the activation distribution and enforce the attention alignment of the $\{P_3, P_4, P_5, P_6\}$ layers of the backbone. The attention map Q^j is calculated by

$$Q^j = \left(F^p_{\text{sum}}\left(A^j\right)\right) \tag{5}$$

where $A^j \in R^{C^j \times H^j \times W^j}$ is the feature map of the given layer $j \in \{P_3, P_4, P_5, P_6\}$ and $F^p_{\text{sum}}(A^j) = \sum_{k=1}^{C^j} |A^j_k|^p$ is a function that collapses the activation tensor A_j in its channel dimension.

The optical information is transferred to the student branch through the KD loss. It can help the student model learn some knowledge that might not learn well in SAR image to enhance the student model. The attention-based KD loss is

$$L_{\text{akd}} = \sum_{j=3}^{6} \left\| Q^j_T - Q^j_S \right\|^2 \tag{6}$$

where Q_T is the attention of teacher network, while Q_S is that of student network.

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Finally, we optimize the teacher–student network with the weighted summation of the detector loss and attention-based KD loss as

$$L_{\text{total}} = L_{\text{det}} + \omega \cdot L_{\text{akd}} \tag{7}$$

where ω is a hyperparameter and we set it to 0.2 in our experiments.

IV. Experiment

A. Dataset and Implementation Details

1) Dataset: We conduct experiments on SpaceNet 6 dataset [59]. This dataset contains 3401 pairs of full-polarized SAR images and corresponding optical images with the spatial resolution of 0.5 m/pixel. Each image has 900 × 900 pixels, in which a total of 820 pairs of images containing oil tanks are selected. Ground truths are marked according to co-registered optical images with bounding boxes. A total of 18 000+ oil tank instances were labeled. Due to the precise geographical registration, the annotation of the optical images can be directly used in the corresponding SAR images. The remaining 2581 pairs of images without oil tanks were used for SSL to improve the performance of SAR feature extractor. According to the georeferenced information, we divided the labeled images into 620 SAR-optical pairs for training, and the rest 200 SAR images for testing, ensuring that the overlap between the train set and test set is as little as possible. The train set can also be used in pretraining.

2) Implementation Details: For the pretraining stage, the ResNet50 network is adopted as the backbone due to its applicability and wide usage. The co-registered optical images and SAR images are fed into the network without augmentation for the proposed OS-SSL. The default augmentation strategy in BYOL [31] is applied when using SAR images for SSL only. The training samples are precropped into 256 × 256 patches or patch pairs before being fed into the network. All the SSL networks are trained for 300 epochs. For the training stage, the teacher and student networks adopt the same structure and are implemented in PyTorch. We conduct our experiments on a platform with 2× NVIDIA RTX3090 GPUs. The network is trained with a stochastic gradient descent optimizer with a learning rate of 0.02 and a batch size of 8. The backbone of the teacher network is pretrained on the ImageNet, while that of the student network is from the pretraining stage. The teacher is pretrained for 24 epochs, while the student is trained for 60 epochs.

3) Evaluation Metrics: In this study, we use the standard MS COCO evaluation metric [61] to assess the performance of oil tank detection, and all the results are reported on the test set. Intersection over union (IoU) is computed between the area covered by the ground-truth bounding box and the predicted bounding box. The true positive (TP) is held true for the IoU above the threshold, while the false positive (FP) is held true for the IoU under the threshold. False negative (FN) is the true object, which is not detected. Then, precision and recall are computed by

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{8}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{9}$$

Then, we sort the precision versus recall by confidence scores to generate the PR curve. Based on the PR curve, the AP scores can be obtained by calculating the area under the PR curve. In this article, we use AP_{50} as the final evaluation metric, which is calculated under the IoU threshold of 0.5.

TABLE I
Quantitative Analysis of Our Proposed Method and State-of-the-Art Methods

	Pre-trained	AP_{50}
Faster R-CNN [21]	ImageNet	45.7
Cascade R-CNN [22]	ImageNet	46.8
DetectoRS [6]	ImageNet	47.1
EO-augmented [48]	ImageNet	47.3
YOLOX [62]	ImageNet	47.3
YOLOv5-r6.1 [60]	ImageNet	**47.6**
Ours(w/o pre-train)	ImageNet	**47.6**
Faster R-CNN [21]	Optical	47.0
Cascade R-CNN [22]	Optical	47.9
DetectoRS [6]	Optical	48.5
EO-augmented [48]	Optical	47.7
YOLOX [62]	Optical	49.4
YOLOv5-r6.1 [60]	Optical	49.6
Ours	OS-SSL	**56.2**

B. Benchmarking Results

1) Quantitative Analysis: We conduct two groups of experiments to compare our method's performance with multiple advanced object detection methods in the case of ImageNet pretraining and optical image-assisted pretraining. We take the Faster R-CNN [21] as the baseline detector to implement our method. In the case of using ImageNet pretraining, we compare several popular methods, including Faster R-CNN [21], Cascade R-CNN [22], and the state-of-the-art detectors DetectoRS [6], EO-augmented [48], YOLOX [62], and YOLOv5-r6.1 [60]. (We choose the "x" version model in the YOLOX and YOLOv5-r6.1 since the "x" version has the highest accuracy. YOLOv5-r6.1 is version 6.1 of YOLOv5, which is published on February 22, 2022). In the case of optical image-assisted pretraining, we use a general pretraining and fine-tuning strategy to train the state-of-the-art detectors for comparison. Specifically, we first pretrain the model on optical images and then use the optical image pretrained model as the initialization to train on the SAR images. All the results are shown in Table I, and note that we maintain the same augmentation strategy for all the experiments for a fair comparison.

It can be observed that when using the ImageNet pretrained backbone to initialize the network, our method can achieve the

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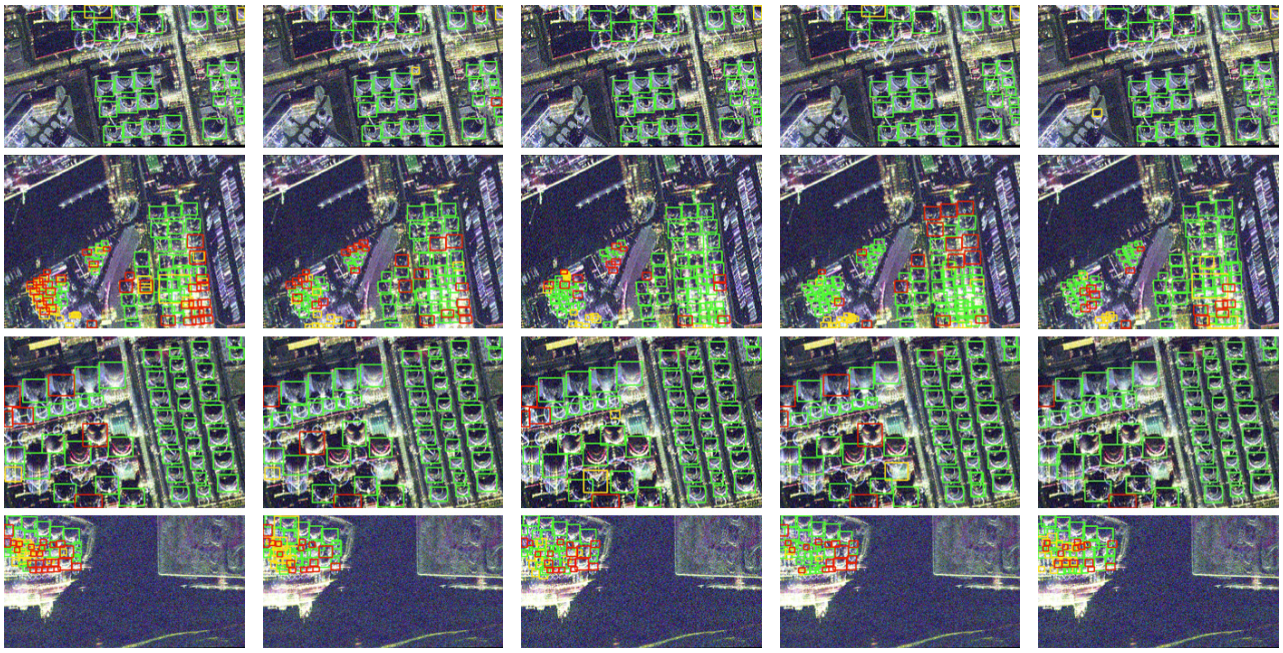


Fig. 5. Visualization of detection results using our proposed method compared with other state-of-the-art methods. The green boxes refer to the correct detection results, the yellow boxes refer to the false detection results, and the red boxes refer to the missing targets. (a) Faster R-CNN [21]. (b) EO-augment [48]. (c) DetectoRS [6]. (d) YOLOv5-r6.1 [60]. (e) Ours.

state-of-the-art performance. Note that "EO-augmented [48]" is originally designed to deal with the building extraction as the semantic segmentation task. Instead, our task here is to detect oil tanks in SAR images. Thus, we apply the corresponding distillation module in Faster R-CNN for the reimplementation. The corresponding co-registered optical images are also used in this method. We can see that using the optical image to guide the SAR image learning is effective and can be transferred to other tasks successfully. Furthermore, when optical images are introduced for pretraining, fine-tuning based on the optical images pretrained model can bring improvements of 0.4–2.3 AP_{50} points. By contrast, our method with the proposed optical-SAR self-supervised pretraining outperforms the baseline by more than nine points. The significant improvement proves that our method can more effectively use optical image information to boost the performance of oil tank detection in SAR images.

The above results show that, compared with the baseline, the advanced methods, such as Cascade R-CNN [22] and DetectoRS [6], can bring a little improvement by using the cascade modules to optimize the object localization. Some improvement can also be achieved through the assistance of optical images knowledge or more advanced backbone and design such as YOLOX [62] and YOLOv5-r6.1 [60]. However, when using the ImageNet pretrained model, the improvement is limited because the amount of SAR training samples for fine-tuning is relatively small, and the network is not sufficient to learn the characteristics of SAR images. When the optical images are used as auxiliary information, simply applying the pretraining and fine-tuning strategy cannot bring significant improvement, where the gain is only about 1–2 points. Moreover, one possible reason for the minor improvement brought by EO-augmented [48] is that the optical image feature has already been utilized for guidance in this strategy before pretraining. In other words, if we further pretrain the model with optical images whose information has already been exploited in feature KD, a further significant improvement will be hard to be obtained. Nonetheless, applying our proposed OS-SSL for pretraining can significantly improve the performance. Compared with the pretraining and fine-tuning, our method provides a new pipeline for using optical images to enhance SAR image learning, which can fully use the information from other modalities and provide a more robust pretrained model.

2) Qualitative Analysis: Fig. 5 presents some qualitative results of our method and other state-of-the-art methods, and the compared methods all use the pretrained model with optical images. The green boxes refer to the correct detection results, the yellow boxes refer to the false detection results, and the red boxes refer to the missing targets. Our method has a higher detection precision and lower missing rate on both dense and sparse oil tanks. Note that when there are red boxes and yellow boxes around the same target, this means that the IoU between the predicted box and the ground-truth box is less than 0.5, indicating that the localization performance of the network is not well. We can observe that our method not only achieves the better quantitative result but also has better localization performance for small and dense oil tanks. The images in the second and fourth rows contain compact oil tanks. The results of other methods all have plenty of false and missing detections, indicating that the localizations of detection results are inaccurate. For the dense and compact oil tanks in SAR images, the scattering interference between oil tanks causes the boundary aliasing in the image, which interferes with the model's localization capability on oil tanks' boundaries.

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TABLE II
COMPONENT ANALYSIS OF THE PROPOSED METHOD

	Baseline	Different Configurations		
AKD		✓		✓
OS-SSL			✓	✓
$AP_{0.50}$	45.7	47.6	54.6	**56.2**

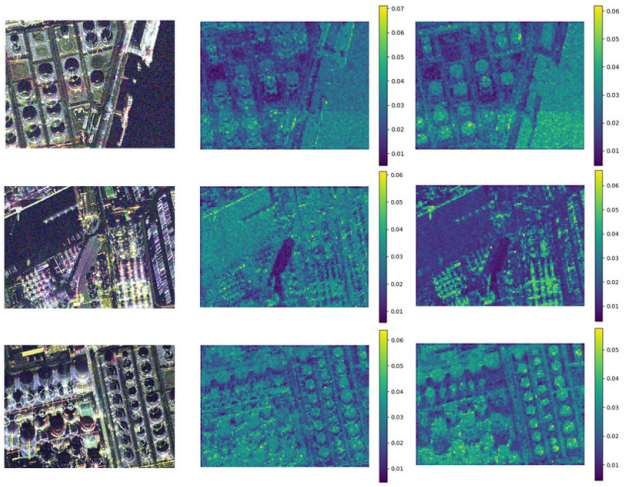


Fig. 6. Visualization of the feature maps extracted from different pretraining models by calculating the sum of squares by compressed channels. The left column is the SAR images, the middle is the feature maps extracted by Imagenet pretrained model, and the right is the feature maps extracted by the cross-modal SSL pretrained model.

Furthermore, dense oil tank areas often contain many smaller objects, and inaccurate positioning always decreases detection precision. Our method optimizes the extracted SAR features in the pretraining stage and introduces the corresponding optical images in the training stage. With the assistance of optical images, the network can further improve the distinction ability in such boundary aliasing conditions and has better localization performance in dense oil tank detection. In reality, oil tanks often exist in this compact arrangement, and our method has a significant improvement in this case. Therefore, it is clear that our method provides more reliable detection than the others.

C. Ablation Study

This work develops oil tank detection solutions from the pretraining backbone and KD perspectives. This section will discuss the impacts of these modules and explore the effects of different settings. Note that Faster R-CNN is used as the baseline detector.

1) Component Analysis: To verify the effectiveness of the proposed two modules individually, we add cross-modal self-supervised pretraining and KD frameworks into the baseline. The results are shown in Table II. On the one hand, the pretrained model by OS-SSL can boost the performance of downstream detectors by a large margin. Nearly, about 9% improvement indicates that the SAR features extracted by the SSL pretrained model are much better than ImageNet pretrained. Fig. 6 shows the comparison of the middle feature map extracted by the SSL pretrained model and the ImageNet pretrained model. As we can see, the feature maps extracted by the SSL pretrained model have a better response to the objects in SAR images, with a smaller response on background. Overall, there is less noise on the SSL pretrained feature maps, which also proves that in the process of cross-modal SSL, the mutual prediction between the optical features and the SAR features can effectively retain the common-share characteristics between the two modalities and suppress the influence of speckle noise in SAR images. On the other hand, KD based on the attention map also results in nearly 2% improvement, indicating that this refined learning can also make the network more robust for SAR oil tank feature extraction.

TABLE III
COMPARISON OF BACKBONE PRETRAINED ON DIFFERENT DATASETS

Methods	Pre-trained Dataset	AP_{50}
Faster R-CNN with ResNet50	No Pre-training	28.6
	ImageNet Pre-trained	45.7
	Optical Pre-trained	47.0
	SAR-only SSL	53.3
	OS-SSL	**54.6**

2) Different Pretrained Models: To verify the influence of different pretrained models, we compare the detectors' performance when using the backbone without pretraining, pretrained on ImageNet, pretrained on the optical oil tank images, and pretrained by proposed OS-SSL. In addition, to verify the advantages of using paired optical images, we also conduct a comparison of the SSL training on SAR images only, and the results are shown in Table III. It can be observed that all the pretrained models can largely improve the network's performance. Meanwhile, as shown in Fig. 7, the better the pretrained model, the faster convergence rate of the network in the early epochs of training. A deep learning network usually works well while training with a large amount of data. When the training samples are limited, we can use pretrained models for model initialization, which can accelerate the convergence speed of the network and improve its performance. As for the SAR image, which is quite different from the ImageNet image, the pretraining on the same modality data is superior to other modalities. The better performance also verifies that the proposed OS-SSL narrows the distance between the extracted SAR and optical image features, and the features of optical images can also be used for reference to further improve the feature extraction of SAR images.

3) Number of Images Used for OS-SSL Pretraining: Studies on SSL show that the number of training samples plays an important role. Therefore, we conduct a set of comparative experiments with different numbers of training samples. In addition, we differentiate whether the training data included oil tank samples or not in this experiment, to verify the influence of oil tank images. As shown in Table IV, we test our method with four groups of different training data on ResNet50. "Images with oil tank" contains 620 pairs of images in the oil tank training set; "Images w/o oil tank" consists of patch pairs cropped from 2581 pairs of images that do

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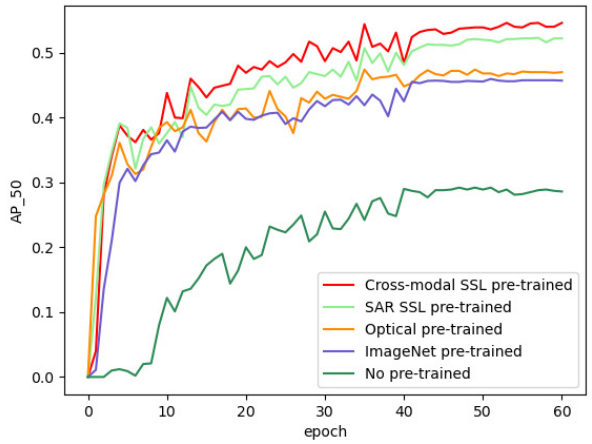


Fig. 7. AP_{50} curve of oil tank detection under different pretraining methods.

TABLE IV
PERFORMANCE OF OIL TANK DETECTION WITH OS-SSL PRETRAINED UNDER DIFFERENT TRAINING SETS

Methods	Pre-training data	AP_{50}
Baseline	ImageNet	45.7
Faster R-CNN with OS-SSL	SN6 images with oil tank	52.6
	SN6 images w/o oil tank	51.9
	Part of SN6 images with oil tank	53.8
	All SN6 images	**54.6**

TABLE V
COMPARISON OF DIFFERENT KD METHODS

Backbone	KD method	AP_{50}
ImageNet Pre-trained	No KD	45.7
	feature maps KD	47.3
	attention maps KD	47.6
OS-SSL Pre-trained	No KD	54.6
	feature map KD	55.3
	attention map KD	56.2

TABLE VI
COMPARISON OF FEATURE MAP LEVELS USED FOR ATTENTION KD

Backbone	P2	P3	P4	P5	P6	AP_{50}
ImageNet Pre-trained						45.7
					✓	46.9
				✓	✓	47.2
			✓	✓	✓	**47.6**
		✓	✓	✓	✓	**47.6**
	✓	✓	✓	✓	✓	46.6
OS-SSL Pre-trained						54.6
					✓	54.9
				✓	✓	55.2
			✓	✓	✓	55.3
		✓	✓	✓	✓	**56.2**
	✓	✓	✓	✓	✓	55.2

not contain oil tank samples; "Part of images with oil tank" contains 620 pairs of images in training set and 1961 pairs of images without oil tank samples; and "All SN6 Images" contains 3201 pairs of all images except the oil tank test set. The results show that the performance of self-supervised training has a close relationship with the number of images used for training, and the more data, the better the performance. Furthermore, when we use the same amount of data with or without oil tank, similar results can be obtained, indicating that the feature extractor can be optimized even without oil tank samples in training data.

4) Different KD Methods: We also apply the ablation study on the cross-modal KD in the second stage and select ResNet50 pretrained on ImageNet and pretrained by cross-modal SSL as the backbone. We compare the baseline detector without KD, using the feature maps for distillation and using the attention maps. Note that when using feature maps for distillation, we calculate the $L2$ loss between the feature maps extracted from the optical image and those of the SAR image. As shown in Table V, the results show that KD based on the attention map has a better performance than distillation using the feature map directly. Since there is an apparent difference between the optical and SAR image, their features should also be different. Therefore, using optical features to directly supervise SAR features is unsuitable, which will lose the specific information in the SAR image. Considering the consistency in the two modalities, the features of the oil tank must be activated in the detection task, which can be reflected in the attention map. Thus, KD based on attention map is more suitable for our task.

5) Choice of Feature Maps for KD: Table VI shows the ablation study of our proposed cross-modal KD under the different levels of feature map. The performance is best when using the P3, P4, P5, and P6 feature map to calculate the attention KD loss. This is because deep features are more inclined to represent the semantic information and more consistent in the different modalities than shallow features. This is why the addition of P2 features does not lead to any improvement.

V. DISCUSSION

Overall, extensive experiments and analyses demonstrate the effectiveness of our proposed method. By comparing with the state-of-the-art methods, we can find that the performance of oil tank detection is limited when using only SAR images. The introduction of information from additional modality data can help more powerful representation learning for SAR images. Compared to optical images, the objects in SAR images usually have inconsecutive contours and severe geometric deformation. At the same time, SAR images also contain severe speckle noise. All these characteristics lead to a weak correlation between the objects' semantics and their appearances, resulting in high uncertainty in semantic parsing and increasing the difficulty of understanding SAR images. Our method provides a feasible solution that combines the scattering intensity information of SAR images and the prior knowledge of other modalities to break through SAR

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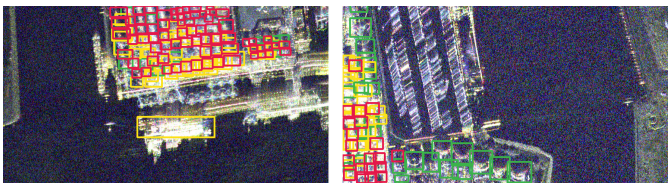


Fig. 8. Visualization of failure cases. The green boxes refer to the correct detection results, the yellow boxes refer to the false detection results, and the red boxes refer to the missing targets.

image learning. Furthermore, unlike the other transfer learning methods, our method performs cross-modal knowledge transfer from both image level and object level. Image-level pretraining using the proposed OS-SSL can provide a better feature extractor for SAR images, which is significant for SAR image understanding based on deep learning. Our method can also be extended to other SAR image target detection tasks.

A. Failure Cases

Fig. 8 reveals the typical failure cases of our method, mainly existing in extremely tiny oil tanks and densely arranged oil tanks. When the densely arranged oil tanks are tiny in the images, they only occupy a few pixels on the feature maps. Therefore, the detector cannot assign enough positive samples to these tiny objects, resulting in insufficient training. Concretely, due to the small size of the oil tanks, a tiny location deviation between the anchor and the ground truth will drastically lower the IoU between them, making the IoU lower than the threshold of positive samples. Most anchors are considered as negative samples, which will lead to insufficient sampling in the training stage and also affect the final accuracy in the inference stage. One of the reasons is that the measurement method based on IoU is not friendly to tiny objects. A straightforward solution is to adaptively zoom in and detect the area of dense and tiny objects [63], [64], but this will introduce a tremendous computational cost. Some other advanced solutions include adopting the measurement robust to tiny objects, such as normalized Wasserstein distance (NWD) [65], a revised Bhattacharyya distance (BD) [66], and seeking balanced training for tiny objects [67]. Except for the effects of the small size, densely arranged tiny objects will make things even worse, i.e., the object detector will be difficult to make optimal label assignments for highly occluded samples, resulting in location errors and false predictions. Meanwhile, nonmaximum suppression (NMS) used in most detectors will suppress adjacent oil tanks when filtering the prediction results, greatly reducing the recall rate. A possible solution is to use transformer-based methods, such as DETR [68] and Deformable-DETR [69], which detect the objects in an end-to-end manner without NMS and can effectively avoid errors.

B. Limitations

Although our study establishes a new pipeline for using optical images to enhance SAR image learning, there are also several limitations. First, our approach requires co-registered optical and SAR images for training. Objects in SAR and optical images are not one-to-one correspondence for unpaired images, and forced alignment of unpaired features will lead to confusion of semantic information learned by the network. It can be alleviated by weak alignment in the feature distribution. Second, our approach only optimizes the representation learning of SAR images from the perspective of image features without fully mining other information such as polarimetric characteristics. In the case of sufficient sensor parameters, some prior knowledge may further improve the interpretation ability of SAR images. These will be further explored in our future work.

VI. Conclusion

In this article, we propose a multistage pipeline for oil tank detection in SAR images enhanced by optical images, which can improve the performance of oil tank detection in SAR images during the inference phase. We employ the deep learning method to deal with the oil tank detection task and introduce optical images to enhance from two aspects. First, in the pretraining stage, an OS-SSL method is adopted to train a feature extractor, which is more suitable for the mode of SAR image. Second, attention-based KD of optical images is used in the detection network training to further learn the optical feature knowledge. Experimental results show that the proposed method can effectively improve the oil tank detection performance in high-resolution SAR images. The ablation study illustrates that the proposed OS-SSL and the attention-based KD improves the accuracy, efficiency, and robustness of oil tank detection in SAR images.

Although we have empirically verified the effectiveness of our approach on the newly built SpaceNet6-OTD dataset and show strong results against state-of-the-art methods for oil tank detection in SAR images, the detection accuracy is still far from that of using clear optical remote sensing images. Nevertheless, we believe that our explorations on paired self-supervised pretraining and optical-SAR KD could inspire further research on this topic.

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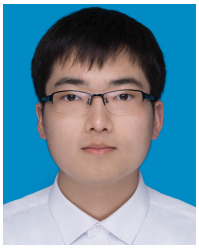
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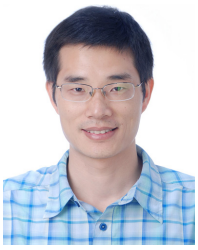
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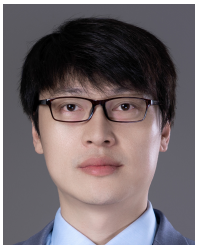
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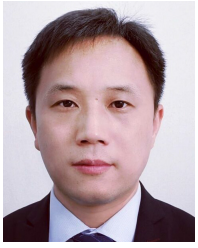
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